

CONTACT INFORMATION	Department of Physics, University of Milano-Bicocca Piazza della Scienza, 3, Milan MI 20126, Italy	E-mail: weichen.wang@unimib.it https://weichenstars.github.io
EDUCATION	Johns Hopkins University , Baltimore MD, United States 9/2016 - 12/2022 Ph. D. in Astrophysics, Department of Physics and Astronomy Thesis Advisor: Susan Kassin (STScI); JHU Advisor: Timothy Heckman Thesis title: <i>Observational studies of the relation between galactic winds and star formation at half the age of the Universe</i> Tsinghua University , Beijing, China 8/2012 - 7/2016 B. Sc. in Physics, Department of Physics Thesis Advisor: Shude Mao	
RESEARCH POSITIONS	University of Milano-Bicocca , Milan MI, Italy 12/2022 - now Postdoc researcher, Department of Physics Advisor: Sebastiano Cantalupo University of California, Santa Cruz CA, United States 2020; 7-9/2015 Visiting student, Department of Astronomy Advisors: Sandra Faber, David Koo	
RESEARCH INTERESTS	Galaxy Formation in the Early Cosmic Epochs Galaxy Formation in Extreme Large-scale Environments Spectroscopic Studies of Galaxies and Their Circumgalactic Medium	
PUBLICATIONS	Four first-author papers published in Nature Astronomy, ApJ, and MNRAS. A total of 17 publications with 1000+ citations in the past 3 years; h-index: 16 See the attached publication list or this ADS link for full publication records.	
AWARDS	Giovani Talenti (Young Talents Award) by U. Milano-Bicocca & Accademia Nazionale dei Lincei of Italy, 2025 The International Astronomical Union Travel Grant, 2019 First-year student award, Johns Hopkins Physics & Astronomy Department, 2016 National Astronomical Observatory of China Scholarship, 2016 Award of excellence by Tsinghua University: 清华学堂计划, 2013-2016	
MEDIA	Press releases by the European Space Agency , Nature Portfolio, U. Milano-Bicocca, Caltech, Swinburne U. (reporting W. Wang+ 2025): <i>"Big Wheel", a surprisingly large disk galaxy discovered in the early universe.</i> News coverage by Scientific American , The Independent, Newsweek, Daily Mail, Il Giornale: <i>JWST Spots Giant Spiral Galaxy Shockingly Early in Cosmic History.</i> News coverage by Phys.org and others (reporting W. Wang+ 2026): <i>'Red Potato' galaxy discovered by astronomers.</i>	

TALKS

Invited Conference Talks:

The Controversial Fate of High- z Galaxy Protoclusters, Spain, 2025

Galaxies and Diffuse Gas in Large-Scale Overdense Environments, Italy, 2024

Department Seminars:

Tsinghua University/Peking University/NAOC, China, 2023/2024

University of Missouri, U.S., 2020

Recent Contributed Talks (Selected):

The Baryon Cycle from Reionization to the Cosmic Noon, Chile, 2025

Tracing Cosmic Evolution with Galaxy Clusters, Italy, 2025

Observing and Simulating Galaxy Evolution in the Era of JWST, Switzerland, 2024

What Matter(s) Around Galaxies 2024, Italy, 2024

Steward/NOIRLab Galaxy Group Lunch Talk, University of Arizona, U.S., 2021

Massively Parallel Large Area Spectroscopy from Space, Portugal, 2021

SERVICE

SOC/LOC member, Conference What Matter(s) Around Galaxies, 2024

Referee for The Astrophysical Journal, Astronomy and Astrophysics

TECHNICAL EXPERTISE

JWST Spectroscopy and Imaging:

Serving as the deputy lead for JWST program GO1835, a spectroscopic survey of more than 50 galaxies in two $z \sim 3$ QSO fields; responsible for the planning, implementation, and analysis of the NIRSpec and NIRCам observations.

Keck spectroscopy: Reducing, organizing, and analyzing the spectra of $\simeq 200$ $z=1$ galaxies as the member of a Keck/DEIMOS program HALO7D (P.I.: GuhaThakurta).

OBSERVATION PROGRAMS (SELECTED)

Summary: 100+ hours on JWST and 120+ hours on VLT & ALMA. Programs joined as P.I. and core member are listed and marked with [P] and [C], respectively.

[C] ALMA Cycle-12 Program (P.I.: A. Pensabene; 27 hours):

The Big Wheel Galaxy: a unique laboratory to study the physics of early disk formation in overdense environments, 2025-2026

[C] JWST Cycle-4 Program (P.I.: B. Wang; 20 hours):

Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking object by hunting for stellar absorption features, 2025-2026

[C] VLT P112 Program (P.I.: S. Cantalupo; 84 hours): *Connecting the dots with MUSE: the Cosmic Web in emission around a massive structure at $z=3$* , 2024-2025

JWST Cycle-2 Program (P.I.: Kassin; 68 hr): *Galaxy angular momentum alignment with filaments at $z\sim 3$: The effect of large scale structure on galaxies*, 2024

[C] JWST Cycle-1 Program (P.I.: S. Kassin; 74 hours): *A Pathfinder for JWST Spectroscopy: Deep High Spectral Resolution Maps of Galaxies over $1 < z < 6$* , 2023

[C] JWST Cycle-1 Program (joined in 2022 with major contributions; P.I.: S. Cantalupo; 24 hours): *Unraveling the Knots of Gaseous Cosmic Web Filaments at $z\sim 3$ through H-alpha Emission Observations*, 2023

ALMA Cycle-8 Program (P.I.: R. Simons): *CO Kinematics at Cosmic Noon: Timing the Redistribution of Metals Around Galaxies*, 23.1 hours, 2022

[P] ALMA Cycle-7 Program (P.I.; 15 hours): *Does molecular gas follow the motion of ionized gas inside typical high-redshift star-forming galaxies?* Observations not completed due to weather and the impact of COVID-19 in Chile, 2021

MENTORSHIP	Xiaohan Wang, Visiting Ph.D student of Tsinghua University, 2025 <i>JWST NIRSpec studies of galaxy metallicity in the high-redshift cosmic web</i> Resulting in one research paper completed & submitted to A&A. M. Francesca Ubaldi, Bicocca undergraduate in physics major, 2024 (thesis project) <i>Relations between galaxy colors and morphology with JWST medium-band imaging</i> Now pursuing Masters in Astronomy degree in Bicocca. Ying Qin, Johns Hopkins undergraduate in physics, 2021-2023 <i>Studying the Mg II emission and ionizing photons from low-mass galaxies at $z \sim 1$</i> Resulting in two conference poster presentations, including one at the AAS Annual Meeting; Now pursuing a career as lawyer in Washington D.C.
TEACHING EXPERIENCE	Volunteer Teaching Assistant, Laboratory of Data Analysis in Astrophysics University of Milano-Bicocca, Spring 2023 Teaching Assistant, General Physics I for Biological Science Majors Johns Hopkins University, Fall 2016 Teaching Assistant, General Physics Laboratory Johns Hopkins University, Fall 2016
OUTREACH ACTIVITIES	Science outreach day at University of Milano-Bicocca, 2024 A community event open to students from local primary/middle schools in Milan; co-designing the exhibition about astrophysical science with JWST Member of the Astro Scholars program, Johns Hopkins University, 2021-2022 An annual program about astrophysics and coding for college students from under-represented backgrounds; serving as a core member of the hiring & education team Member of the Physics and Astronomy Graduate Students Outreach Team (PAGS), Johns Hopkins University, 2017-2019 Supporting visits of students from Baltimore local primary/middle schools around once per semester and teaching fundamental physics with educational demos Volunteer teacher at the Pengzhai Primary School, Guizhou, China, Summer 2013 Teaching multiple STEM-related courses for Grade 3-6; the school, with very limited resources, is located in one of the least developed areas of the country.

OVERVIEW	Four first-author papers published in Nature Astronomy, ApJ, and MNRAS A total of 17 publications with 1000+ citations in the past 3 years; h-index: 16 Full records are also available on the NASA ADS via this link. Research topic categories marked below: C1: Early-epoch galaxy formation in extreme cosmic environments; C2: Interplay between galaxies and the cosmic gas; C3: Galaxy star formation and dust.
FIRST-AUTHOR PAPERS	[5,C1] W. Wang, S. Cantalupo, M. Galbiati, et al. arXiv: 2601.20473, submitted to A&A (2026): <i>A Quiescent Galaxy in a Gas-Rich Cosmic Web Node at $z \sim 3$</i> [4,C1] W. Wang, S. Cantalupo, A. Pensabene, et al. Nature Astron., 9, 710 (2025): <i>A Giant Disk Galaxy Two Billion Years After The Big Bang</i> (Editorial Highlight) [3,C2] W. Wang, S. A. Kassin, S. M. Faber, D. C. Koo, et al., ApJ, 930, 146 (2022): <i>The Baltimore Oriole's Nest: Cool Winds from the Inner and Outer Parts of a Star-Forming Galaxy at $z = 1.3$</i> [2,C3] W. Wang, S. A. Kassin, C. Pacifici, et al., ApJ, 869, 161 (2018): <i>Galaxy Inclination and the IRX-β Relation: Effects on UV Star Formation Rate Measurements at Intermediate to High Redshifts</i> [1,C3] W. Wang, S. M. Faber, F.-S. Liu, et al., MNRAS, 469, 4063 (2017): <i>UVI colour gradients of $0.4 < z < 1.4$ star-forming main-sequence galaxies in CANDELS: dust extinction and star formation profiles</i> Publications [1-3] are all referenced by recent review articles on the Annual Review of Astronomy and Astrophysics (ARA&A).
FIRST-AUTHOR PAPERS IN PREP.	[1,C2] W. Wang, S. A. Kassin, S. M. Faber, D. C. Koo, et al., in prep.: <i>The Roles of Integrated and Resolved Galaxy Star Formation in Driving Winds at $z \sim 1$</i>; Draft completed and to be submitted to ApJ in Spring 2026.
PAPERS WITH LEADING CONTRIBUTIONS	*[2,C1] X. Wang, S. Cantalupo, W. Wang, et al. submitted to A&A (2025): <i>Metal enrichment of galaxies in a massive node of the Cosmic Web at $z \sim 3$</i> [1,C1] A. Pensabene, S. Cantalupo, W. Wang, et al. A&A, 701, A120 (2025): <i>ALMA survey of a massive node of the Cosmic Web at $z \sim 3$: II. A dynamically cold disk galaxy in the proximity of a hyperluminous quasar</i> *paper led by PhD student co-mentored
CO-AUTHOR PUBLICATIONS	[23] Song et al., ApJ, 997, 126 (2026): <i>The Size Evolution and the SizeMass Relation of Ly Emitters across $3 \lesssim z \lesssim 7$ as Observed by JWST</i> [22] Galbiati et al., A&A, 696, A95 (2025): <i>Connecting the growth of galaxies to the large-scale environment in a massive node of the Cosmic Web at $z \sim 3$</i> [21] de la Vega et al., ApJ, 980, 168 (2025): <i>Improved SED-fitting Assumptions Result in Inside-out Quenching at $z \sim 0.5$ and Quenching at All Radii Simultaneously at $z \sim 1$</i> [20] Travascio et al., A&A, 694, A165 (2025): <i>X-ray view of a massive node of the</i>

Cosmic Web at $z \sim 3$: I. An exceptional overdensity of rapidly accreting supermassive black holes

- [19] Liang, et al., **ApJ**, 986, 60 (2025): *Cosmic Himalayas: The Highest Quasar Density Peak Identified in a 10,000 deg² Sky with Spatial Discrepancies between Galaxies, Quasars, and IGM H I*
- [18] Finkelstein et al., **ApJL**, 983, L4 (2025): *The Cosmic Evolution Early Release Science Survey (CEERS)*
- [17] Ren et al., **ApJ**, 982, 200 (2025): *The Evolution of the Size and Merger Fraction of Submillimeter Galaxies across $1 < z < 6$ as Observed by JWST*
- [16] Pensabene et al., **A&A**, 684, A119 (2024): *ALMA survey of a massive node of the Cosmic Web at $z \sim 3$. I. Discovery of a large overdensity of CO emitters*
- [15] Li et al., **ApJS**, 275, 27 (2024): *MAMMOTH-Subaru. II. Diverse Populations of Circumgalactic Ly α Nebulae at Cosmic Noon*
- [14] Wang et al., **MNRAS**, 533, 2026 (2024): *SDSS-IV MaNGA: stellar rotational support in disc galaxies versus central surface density and stellar population age*
- [13] de Beer et al., **MNRAS**, 526, 1850 (2023): *Resolving the physics of quasar Ly α nebulae (RePhyNe): I. Constraining quasar host halo masses through circumgalactic medium kinematics*
- [12] Pérez-González et al., **ApJL**, 946, L16 (2023): *CEERS Key Paper. IV. A Triality in the Nature of HST-dark Galaxies*
- [11] Guo et al., **ApJL**, 945, L10 (2023): *First Look at $z > 1$ Bars in the Rest-frame Near-infrared with JWST Early CEERS Imaging*
- [10] Pacifici et al., **ApJ**, 944, 141 (2023): *The Art of Measuring Physical Parameters in Galaxies: A Critical Assessment of Spectral Energy Distribution Fitting Techniques*
- [9] Zavala et al., **ApJL**, 943, L9 (2023): *Dusty Starbursts Masquerading as Ultra-high Redshift Galaxies in JWST CEERS Observations*
- [8] Finkelstein et al., **ApJL**, 940, L55 (2022): *A Long Time Ago in a Galaxy Far, Far Away: A Candidate $z \sim 12$ Galaxy in Early JWST CEERS Imaging*
- [7] Pharo et al., **ApJS**, 261, 12 (2022): *The Dwarf Galaxy Population at $z \sim 0.7$: A Catalog of Emission Lines and Redshifts from Deep Keck Observations*
- [6] Simons et al., **ApJ**, 874, 59 (2019): *Distinguishing Mergers and Disks in High-redshift Observations of Galaxy Kinematics*
- [5] Liu et al., **ApJ**, 860, 60 (2018): *On the Transition of the Galaxy Quenching Mode at $0.5 < z < 1$ in CANDELS*
- [4] Jiang et al., **ApJ**, 854, 70 (2018): *The Isophotal Structure of Star-forming Galaxies at $0.5 < z < 1.8$ in CANDELS: Implications for the Evolution of Galaxy Structure*
- [3] Guo et al., **ApJ**, 853, 108 (2018): *Clumpy Galaxies in CANDELS. II. Physical Properties of UV-bright Clumps at $0.5 < z < 3$*
- [2] Liu et al., **ApJL**, 844, L2 (2017): *The Origins of UV-optical Color Gradients in Star-forming Galaxies at $z \sim 2$: Predominant Dust Gradients but Negligible sSFR Gradients*
- [1] Liu et al., **ApJL**, 822, L25 (2016): *The UV-Optical Color Gradients in Star-forming Galaxies at $0.5 < z < 1.5$: Origins and Link to Galaxy Assembly*